CIVE97129
Freight Transport

Assignment 1

Imperial College London

Supply Chain Game Introduction

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Transport Systems & Logistics Laboratory

Introduction

SUPPLY CHAIN GAME

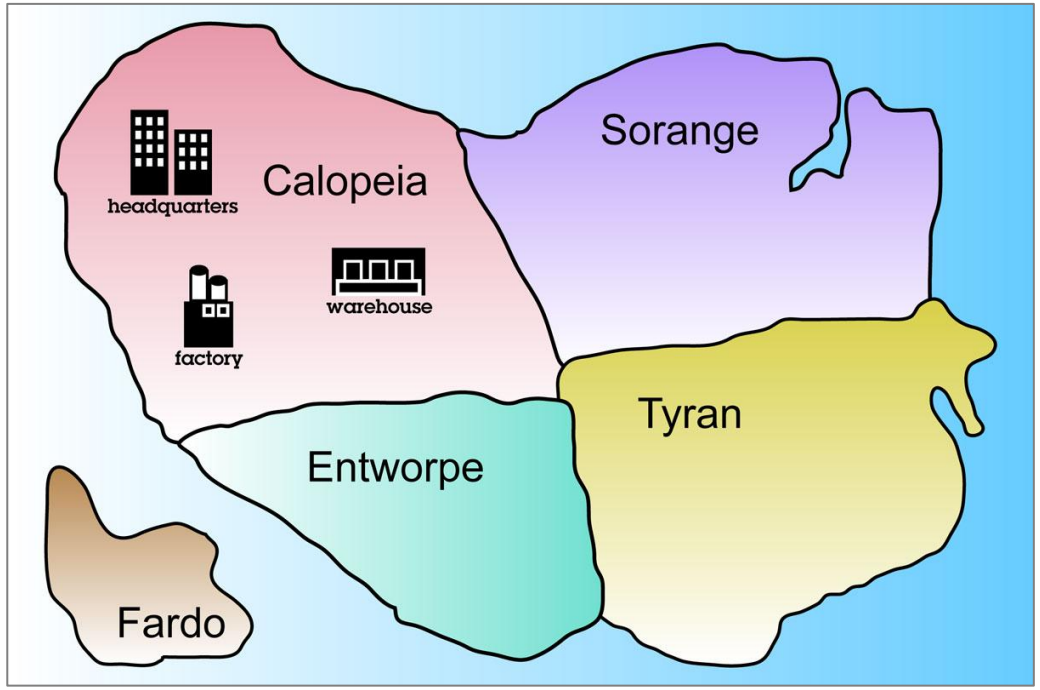
- Web-based discrete event simulator of a network of factories and warehouses.
- There is one single firm producing a single product: drums of an industrial chemical.
- Student teams must manage the company's supply chain in the fictional continent of Pangea and an adjacent island called Fardo.
- The game has two modes: One Region game and Network game.



Overview

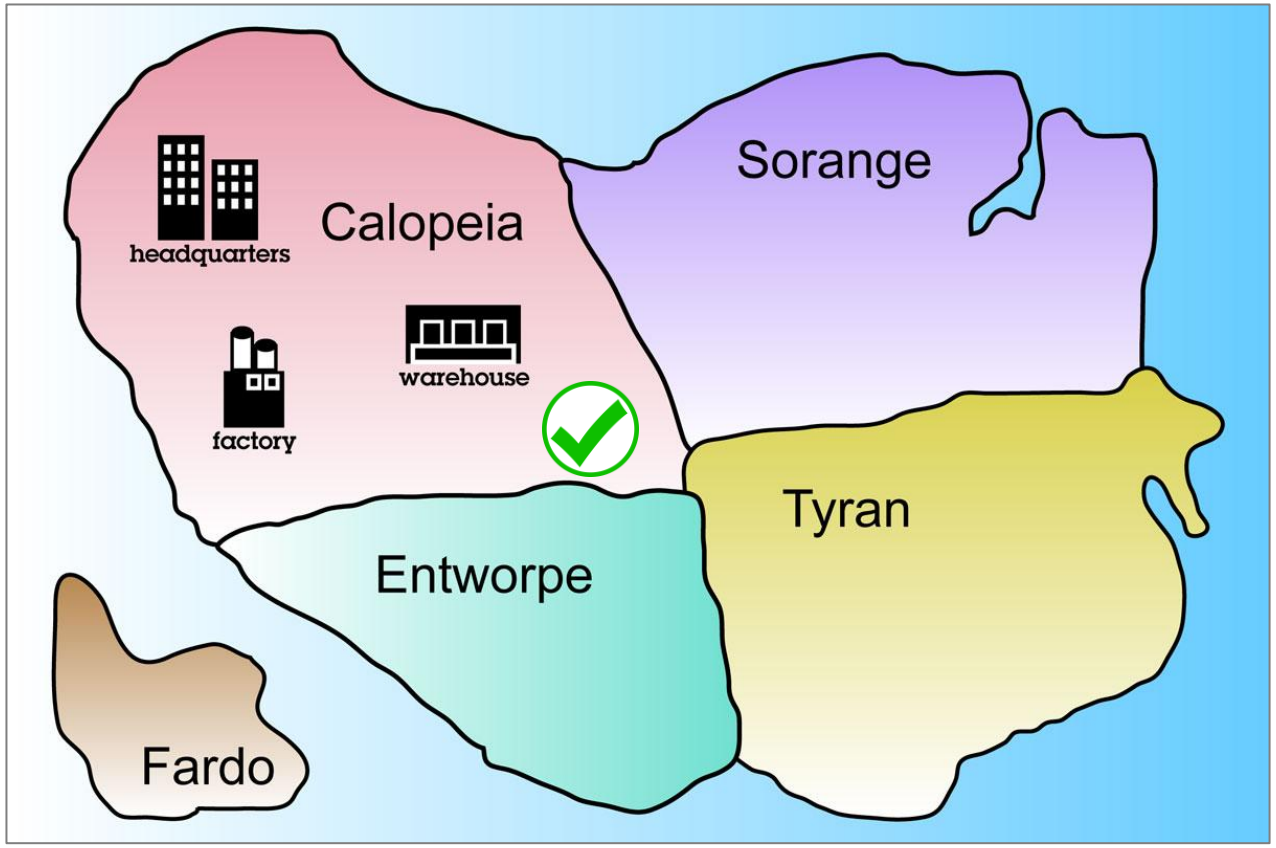
SUPPLY CHAIN GAME

- The continent has four distinct regions: Calopeia, Sorange, Entworpe and Tyran. In addition, there is a small island to the south west called Fardo.
- In the One Region mode, Calopeia is the only region available.
- The factory produces batches of the chemical and sends them to the warehouse.
- Headquarters just hold relevant information regarding demand and cash balance.



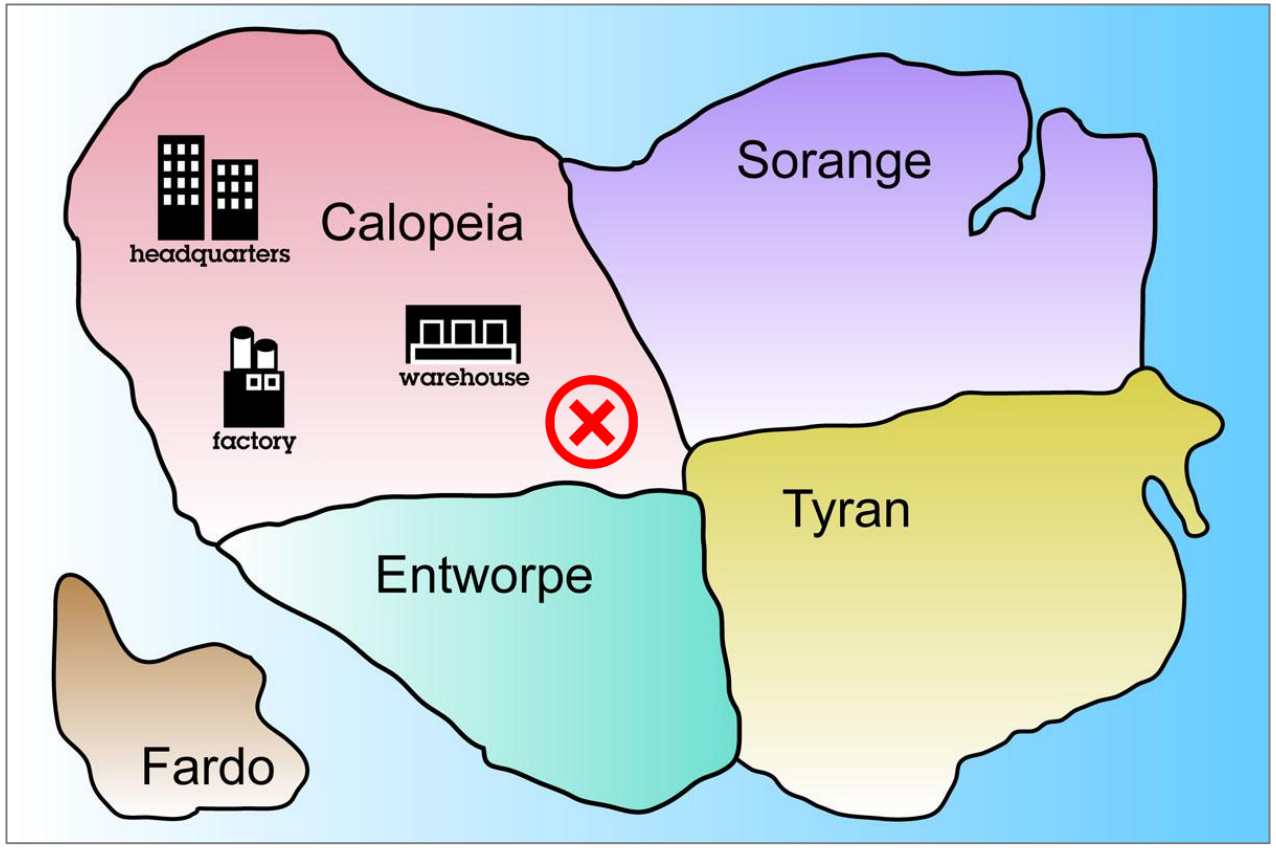
SUPPLY CHAIN GAME

- Orders are filled from the warehouse as soon as they are available.



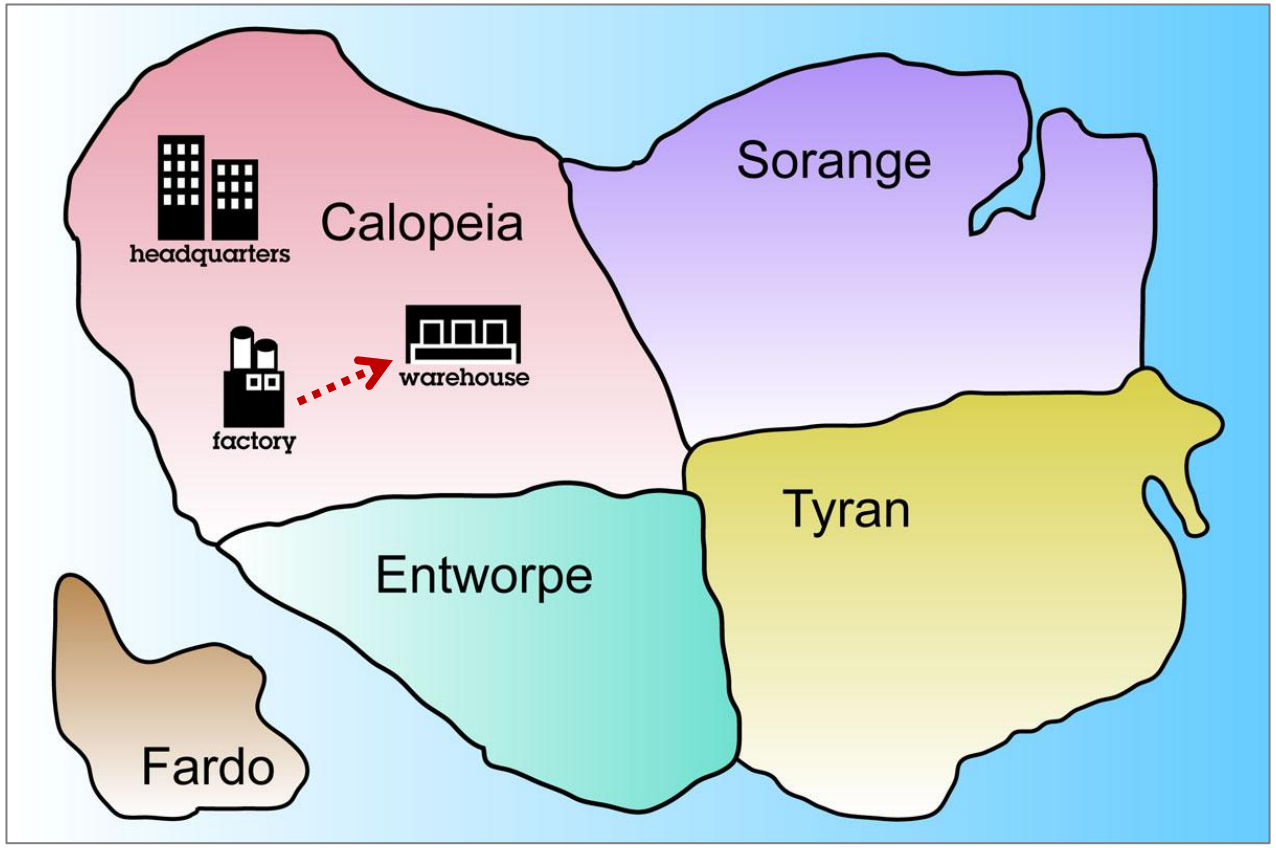
SUPPLY CHAIN GAME

- Orders taking longer than 1 day to fill are lost to competitors.



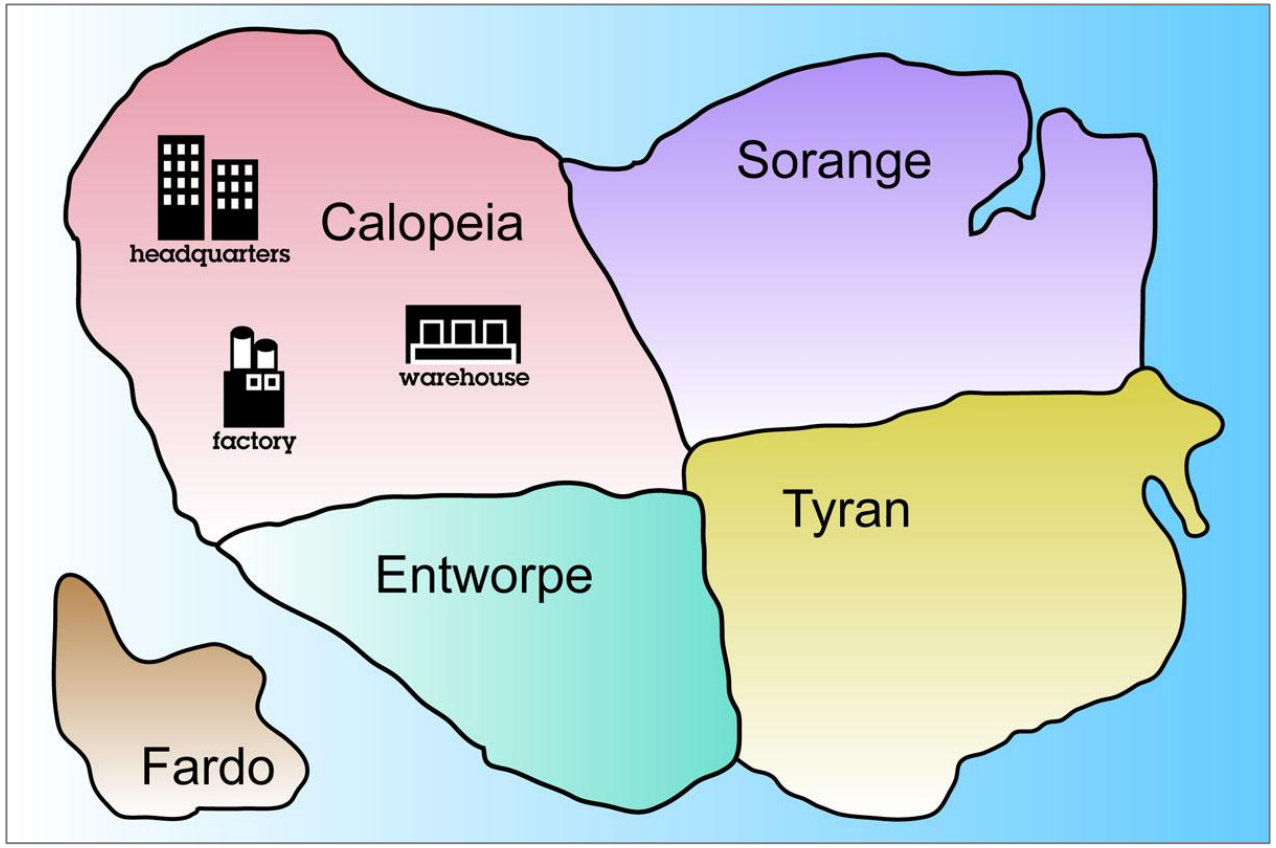
SUPPLY CHAIN GAME

- The warehouse pulls production from the factory.



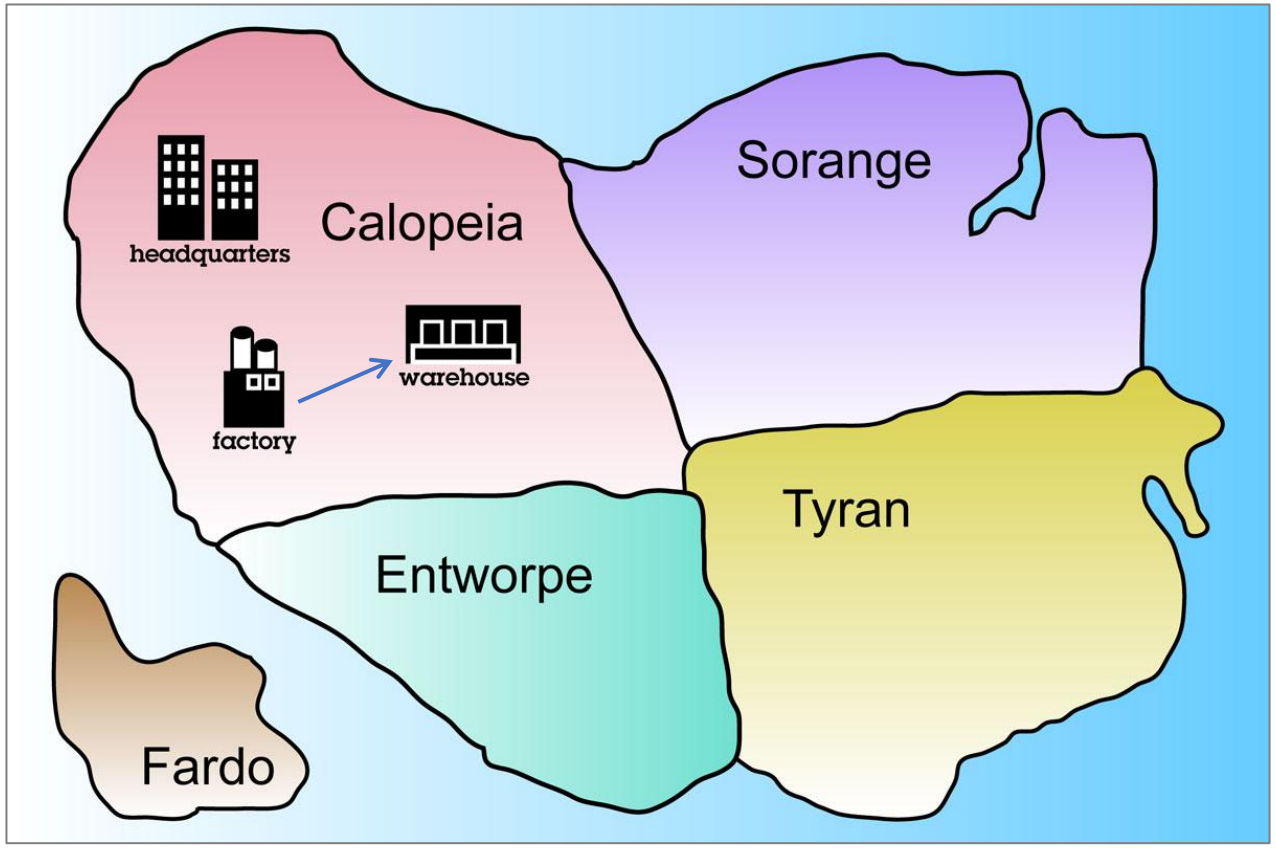
SUPPLY CHAIN GAME

- The factory produces one batch of product at a time and completes each batch all at once.



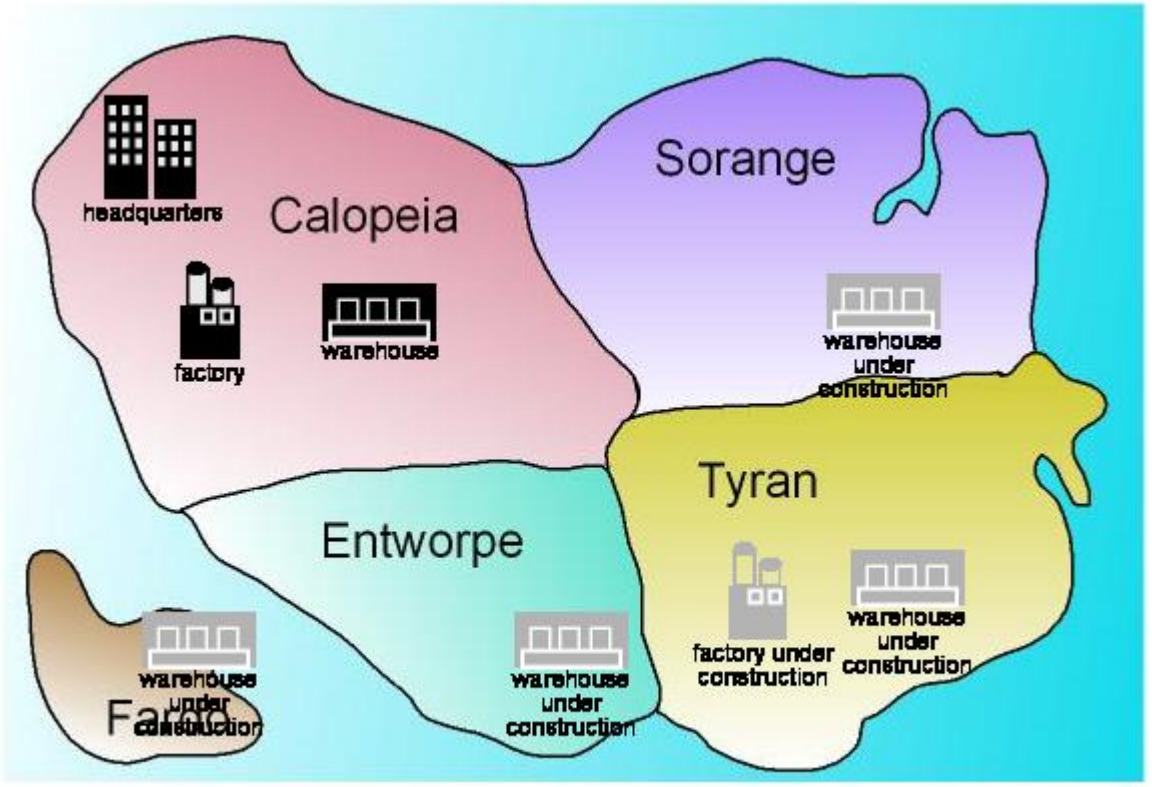
SUPPLY CHAIN GAME

- After each batch is completed, it can be shipped to the warehouse by mail or by truck.



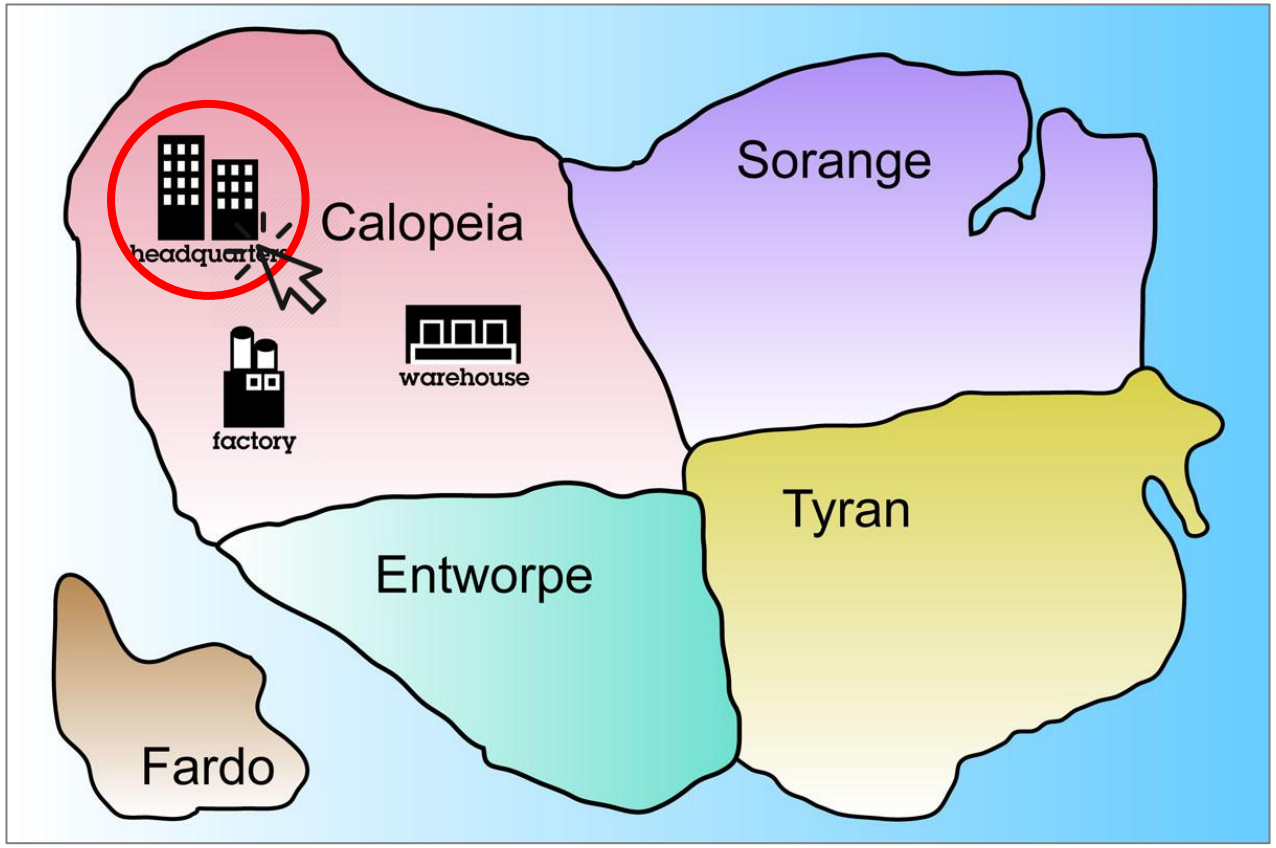
SUPPLY CHAIN GAME

- In the lower menu, you can display the current score, decision history and cash balance.



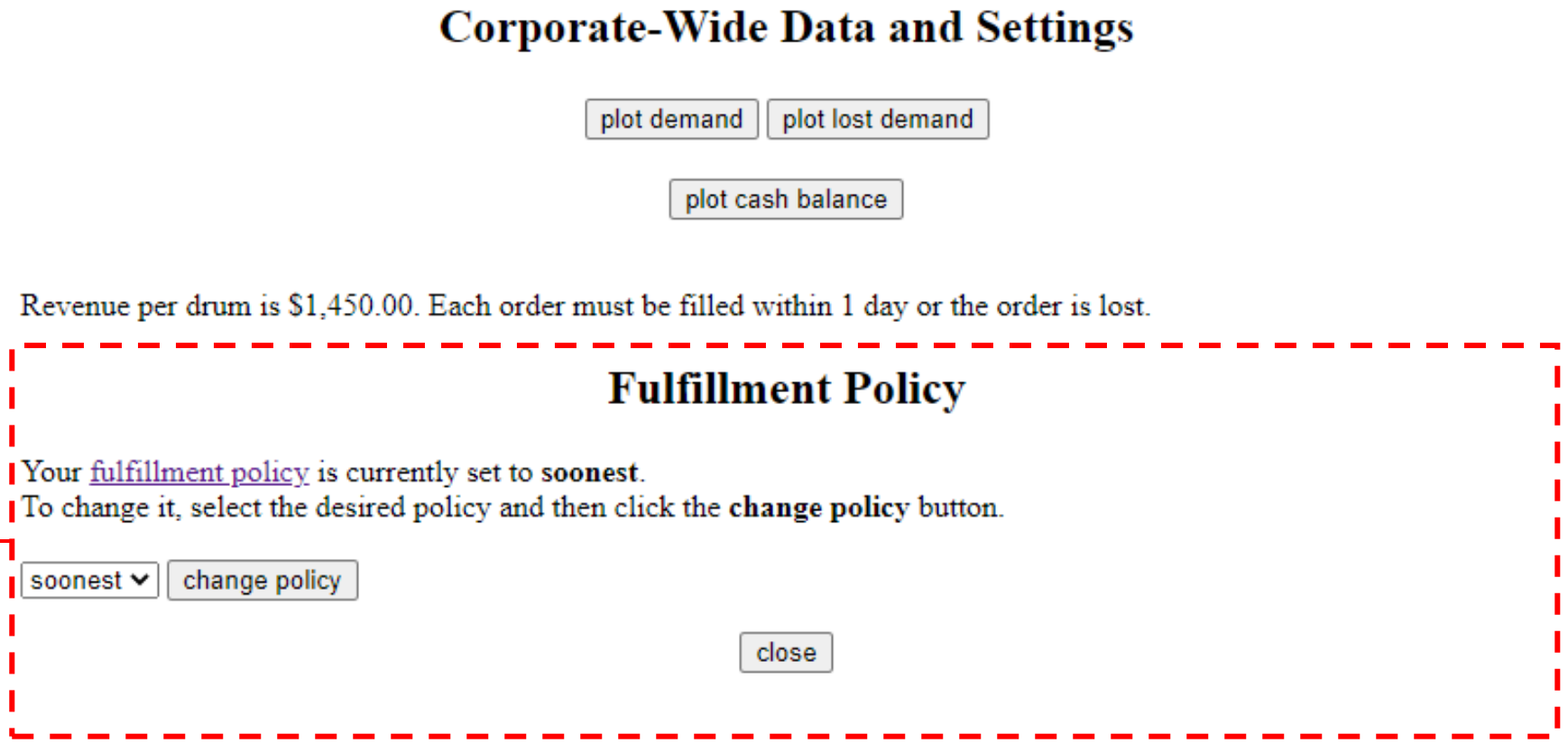
SUPPLY CHAIN GAME

- Click in headquarters to see the past demand, lost demand and cash balance.



SUPPLY CHAIN GAME

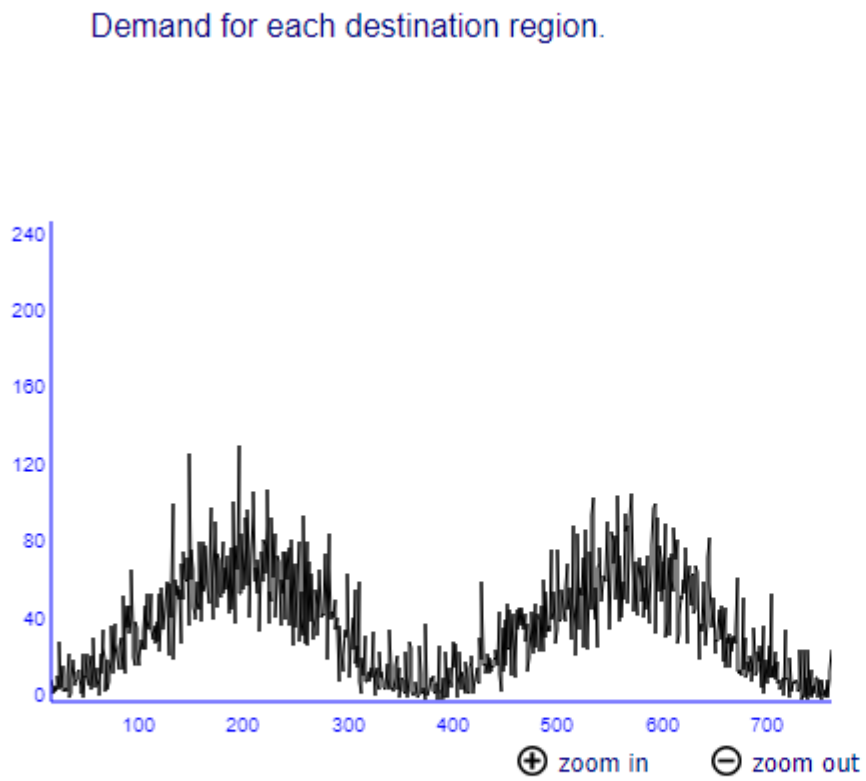
- Click inside ‘plot demand’ to see historic demand levels for all regions.

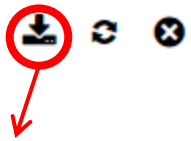


Only available in Network Game

SUPPLY CHAIN GAME

- You can see the demand data for each region and download an excel file.





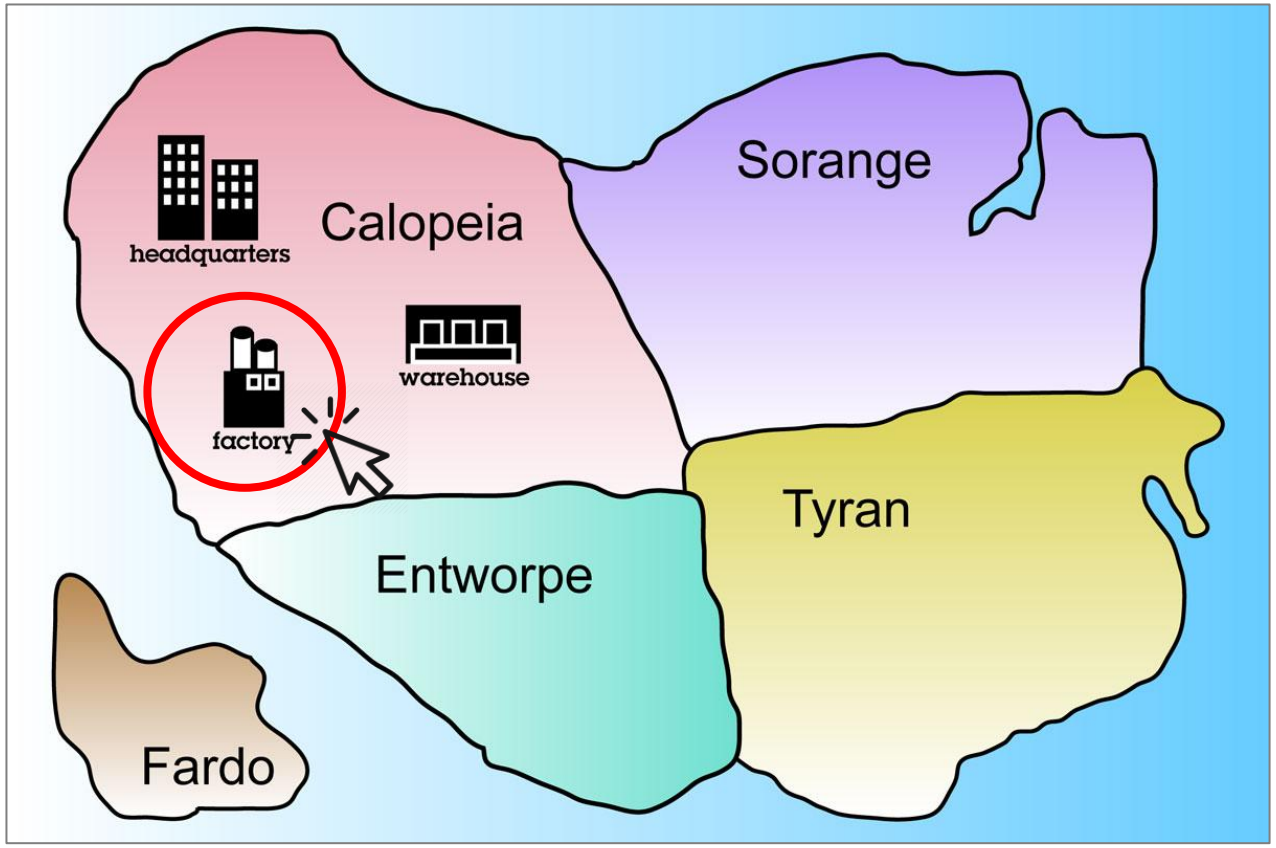
Download data in excel format



See data in table format

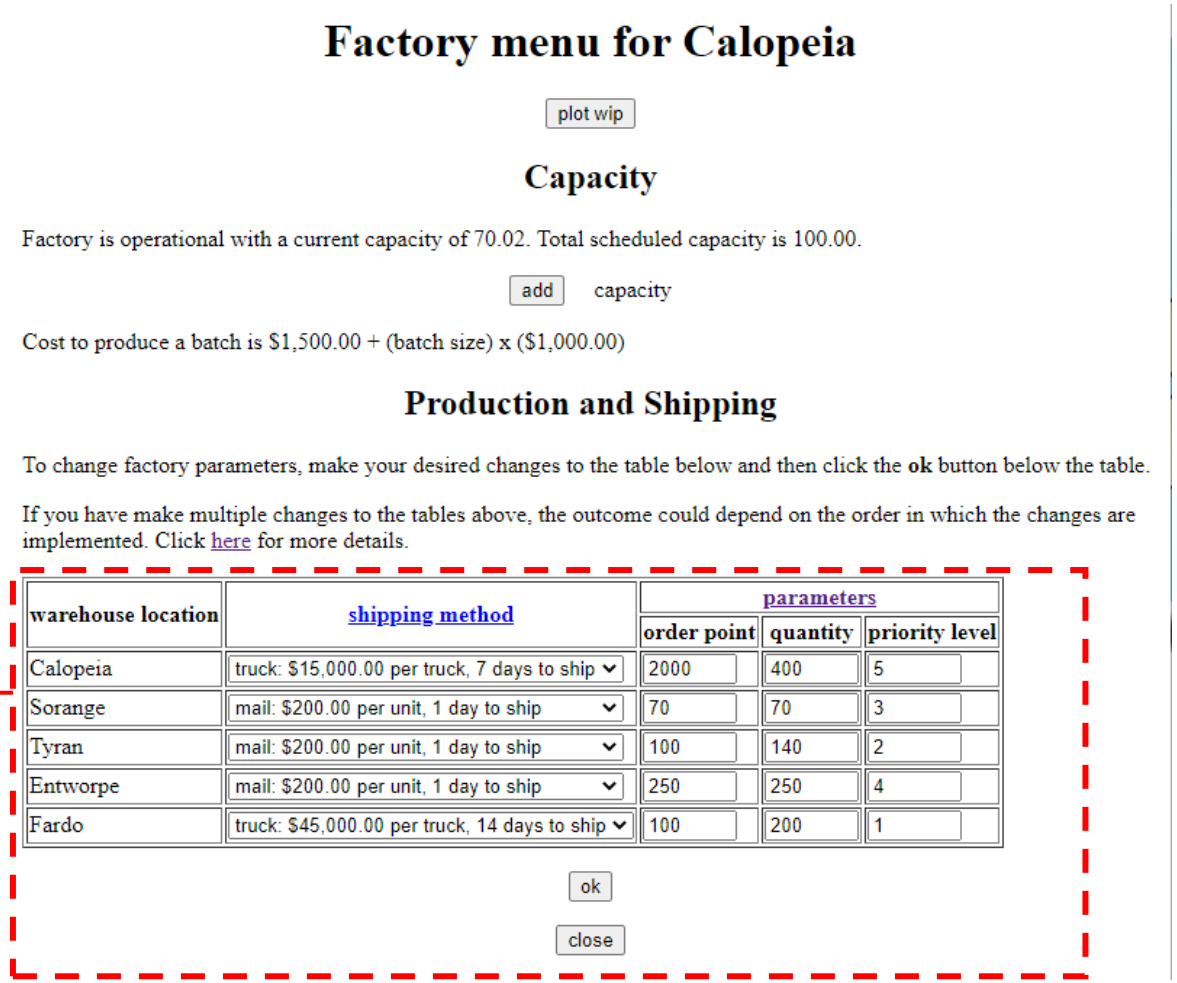
SUPPLY CHAIN GAME

- Click in factory to see the Work in Progress (WIP) and to add capacity (you are not allowed to remove capacity)



SUPPLY CHAIN GAME

- Inside the factory menu you can add production capacity. The cost of adding a single drum of capacity is $50000. Note that this decision will come to effect in 90 days, so try to plan in advance.



Factory menu for Calopeia

plot wip

Capacity

Factory is operational with a current capacity of 70.02. Total scheduled capacity is 100.00.

add capacity

Cost to produce a batch is $1,500.00 + (batch size) x ($1,000.00)

Production and Shipping

To change factory parameters, make your desired changes to the table below and then click the **ok** button below the table.

If you have make multiple changes to the tables above, the outcome could depend on the order in which the changes are implemented. Click here for more details.

warehouse location	shipping method	parameters		
		order point	quantity	priority level
Calopeia	truck: $15,000.00 per truck, 7 days to ship	2000	400	5
Sorange	mail: $200.00 per unit, 1 day to ship	70	70	3
Tyran	mail: $200.00 per unit, 1 day to ship	100	140	2
Entworpe	mail: $200.00 per unit, 1 day to ship	250	250	4
Fardo	truck: $45,000.00 per truck, 14 days to ship	100	200	1

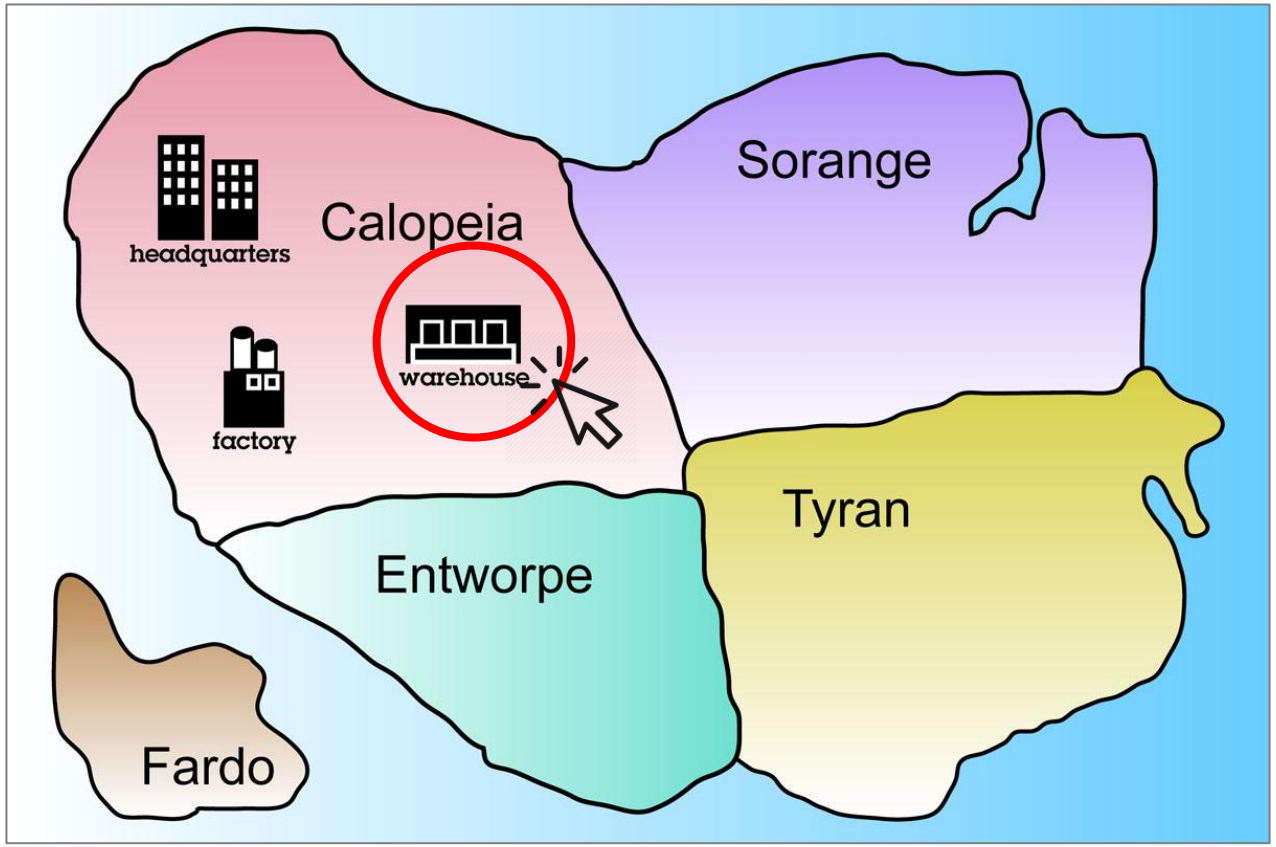
ok

close

Only available in Network Game

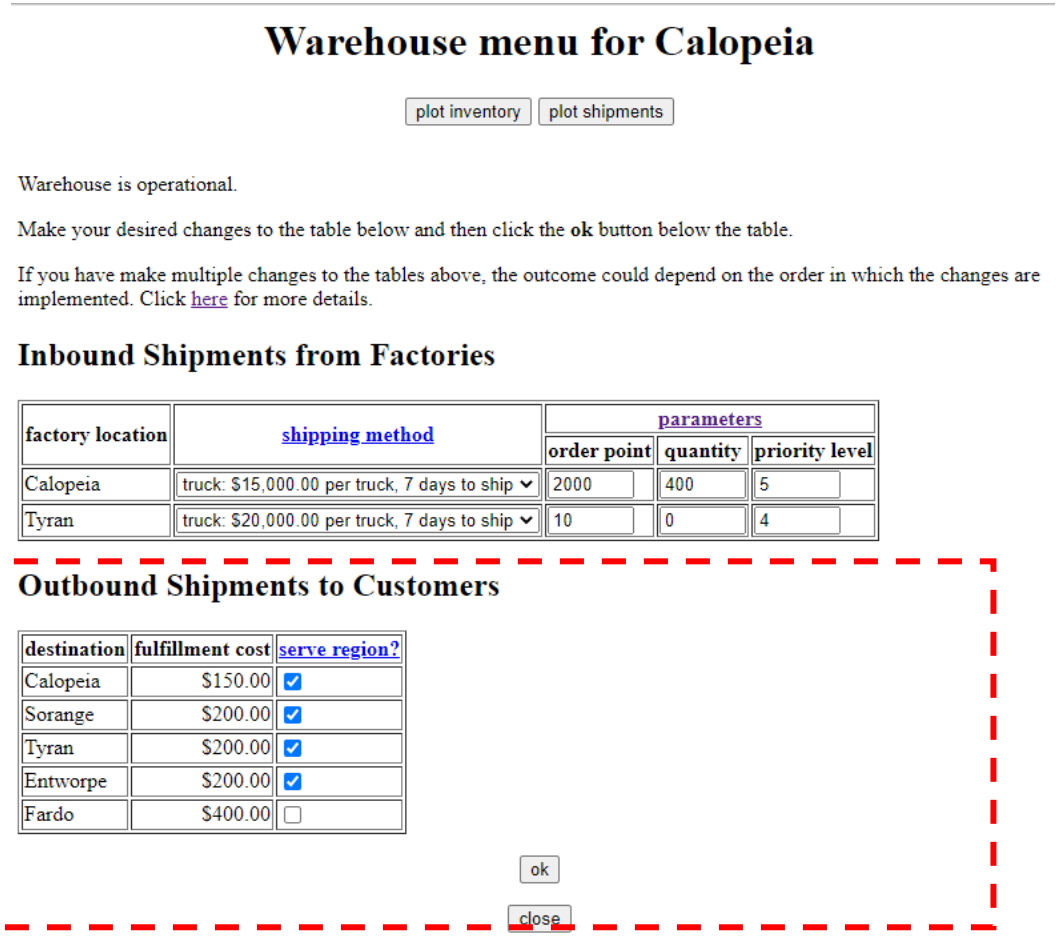
SUPPLY CHAIN GAME

- Click in warehouse to see the Inventory and shipments data/plots.



SUPPLY CHAIN GAME

- Inside the warehouse menu you can also set the order point, quantity and priority level for each region (Network Game).



Warehouse menu for Calopeia

plot inventory | plot shipments

Warehouse is operational.

Make your desired changes to the table below and then click the **ok** button below the table.

If you have make multiple changes to the tables above, the outcome could depend on the order in which the changes are implemented. Click here for more details.

Inbound Shipments from Factories

factory location	shipping method	parameters		
		order point	quantity	priority level
Calopeia	truck: $15,000.00 per truck, 7 days to ship	2000	400	5
Tyran	truck: $20,000.00 per truck, 7 days to ship	10	0	4

Outbound Shipments to Customers

destination	fulfillment cost	serve region?
Calopeia	$150.00	☑
Sorange	$200.00	☑
Tyran	$200.00	☑
Entworpe	$200.00	☑
Fardo	$400.00	☐

ok

close

Only available in Network Game

Finance

REVENUE

- Product revenue: $1450 per drum
- Interest: 10% per year, compounded daily



EXPENSE

- Production cost: $2000 + $900 x batch size
- Shipping cost: $150 per drum for mail or $15,000 per truck (one truck can hold up to 200 drums)
- Finished goods holding cost: $90 per drum per year
- Customer fulfilment costs: $160 per drum

Decisions and Objective

DECISIONS

- Add capacity to the factory (cannot be reduced)
- Determine factory batch size (order quantity)
- Set reorder point to trigger orders from the factory
- Select shipping mechanism: mail or truck
- Network Run: install new factories and warehouses in the different regions.

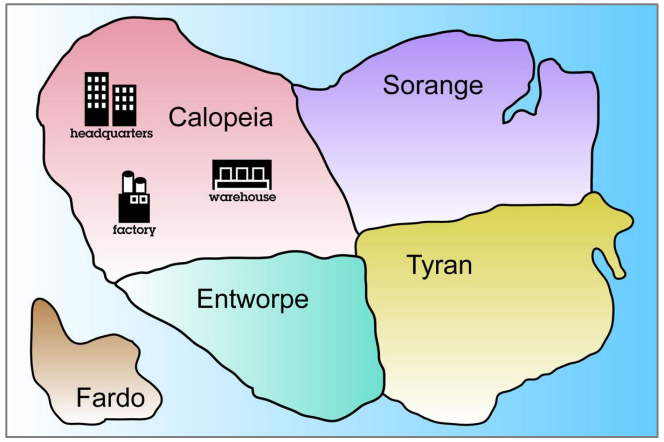
OBJECTIVE

- The objective is to maximise the company's cash balance over two years of simulation. Simulation starts at day 730 and ends at day 1460.

Stages

STAGE 1: ONE REGION TRIAL RUN

- This first run will be for you to familiarise with the simulation framework. It will be accelerated.

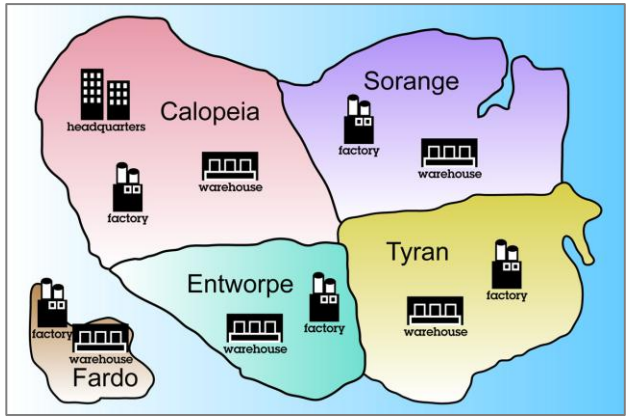


STAGE 2: ONE REGION PROPER RUN

- This will be a proper run and will last for a week.

STAGE 3: NETWORK RUN

- Last stage will be the full network run and will also last for a week.
- You can make additional decisions to install factories and warehouses in every region.

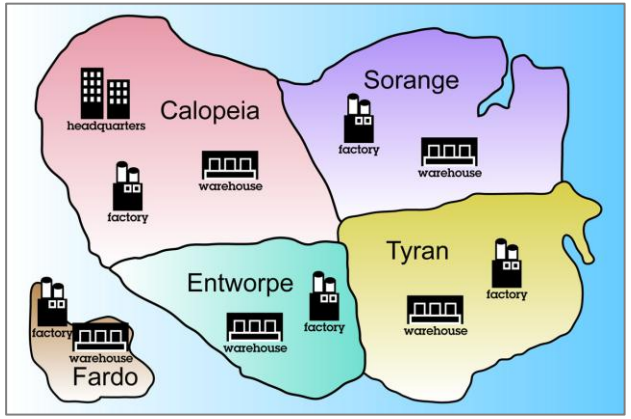


STAGE 3: NETWORK RUN

- Each of the 5 regions has their own demand and can be supplied from any of the other 4 regions.

- You will have to decide if it is worth to install factories and warehouses in other regions given the different shipment costs, shipment times and fulfilment costs for each region.

- You will also have to decide order points, quantities and priorities for each region.



STAGE 3: NETWORK RUN

- With this additions, you will now have to decide on which factories supply each region and the corresponding shipping method.
- Also, there is a priority level, which establishes which regions should be supplied first. 5=highest priority, 1=lowest priority.

Production and Shipping

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Fardo	truck: $45,000.00 per truck, 14 days to ship	100	200	1

STAGE 3: NETWORK RUN

Parameters:

- Cost of building an additional factory: $500,000 + $50,000*(capacity). 90 days to build.
- Cost of building a warehouse: $100,000. 60 days to build.
- Transportation costs and times:

Origin and Destination	Cost per Truckload	Cost to Mail One Drum
Same region	$15,000	$150
Different regions on continent	$20,000	$200
Between continent and Fardo	$45,000	$400

Origin and Destination	Truck	Mail
Same region	7 days	1 day
Different regions on continent	7 days	1 day
Between continent and Fardo	14 days	2 days

- Fulfilment costs: same as mail.

STAGE 3: NETWORK RUN

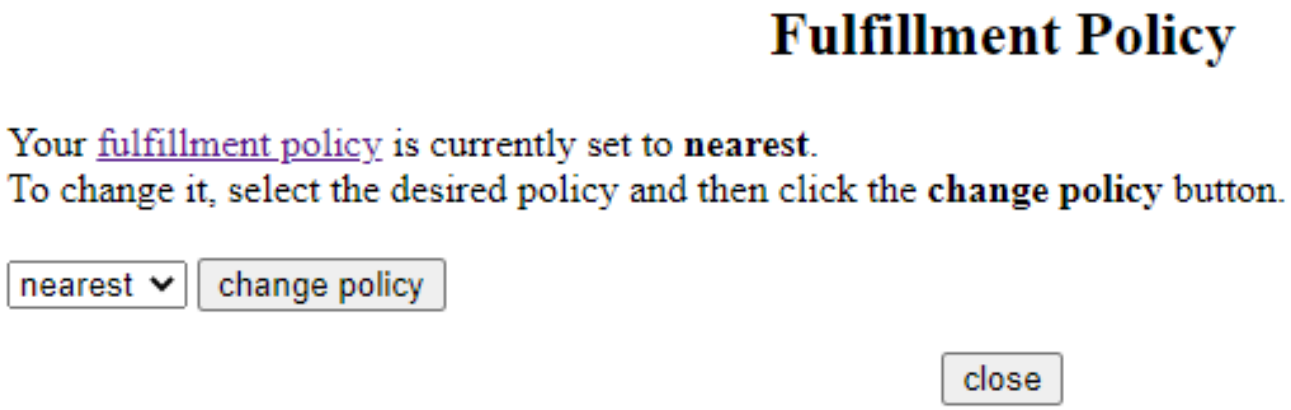
Demands:

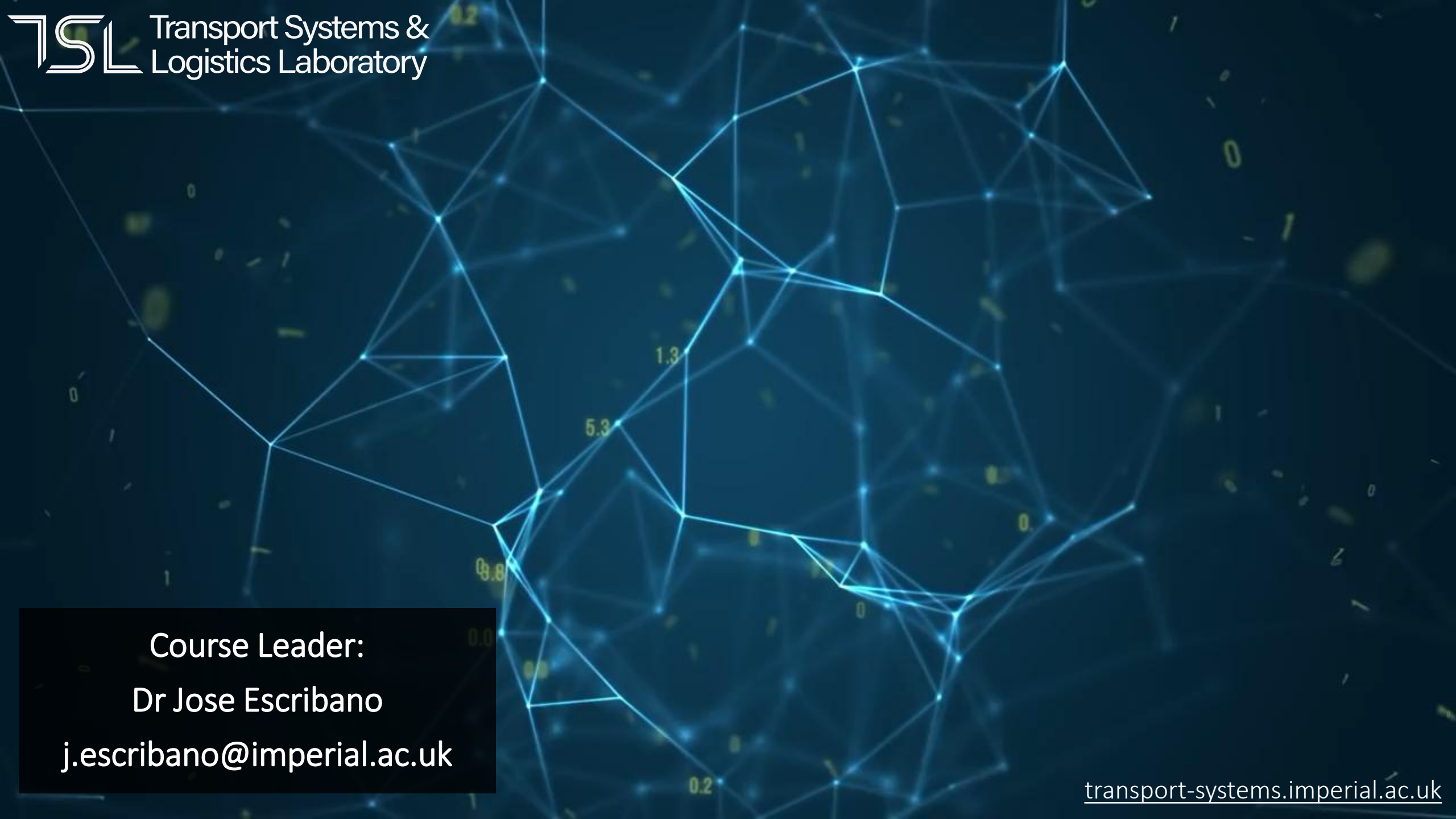
- Calopeia: demand is highly seasonal but otherwise very stable.
- Sorange: the long run average will grow linearly until day 1430.
- Tyran and Fardo: demand began at day 640 and grew to its final long-run average one-month later.
- Entworpe: Demand began at day 640. Order size is fixed at 250 units, but they arrive at random. Not seasonal and not trending upward or downward.
- All Demands will decrease linearly to zero after day 1430.

STAGE 3: NETWORK RUN

- In the headquarters section, there is also an additional variable: Fulfilment Policy

- If there are multiple warehouses, then it may be possible for a customer order to be fulfilled by more than one warehouse when the order arrives. In that case, the fulfilment policy determines which warehouse fulfils the order. There are two possible policies: soonest and nearest.





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